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Atmosphere-driven compound extremes: record marine heatwave and phytoplankton bloom in the Ross Sea during summer 2022–2023

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Abstract

We document an extreme marine heatwave (MHW) in the eastern Ross Sea during the 2022–2023 austral summer (DJF), the most intense thermal event in the satellite record. Using satellite-retrieved sea surface temperature (SST), ocean colour, sea ice, and atmospheric-oceanic reanalysis data, we show that the eastern Ross Sea experienced a Category IV (extreme) MHW during this season, with the regional-mean cumulative intensity reaching 127.22 °C·days in January 2023 and summer-mean SST anomalies locally peaking at approximately +2.5 °C above the climatological mean. Summer-mean chlorophyll-a (Chl-a) concentration reached 3.67 mg m^{−3}, the highest value observed in the eastern Ross Sea since satellite ocean-colour records began in 1997. The ecological response followed a threshold-dependent sequence: early sea-ice retreat initially removed light limitation, followed later by a strong Chl-a increase as sustained warming and upper-ocean restratification favoured bloom development. In contrast, persistent ice cover in Terra Nova Bay prevented a comparable thermal or biological response. A meridional wind dipole drove divergent ice transport and the early expansion of the eastern Ross Sea polynya, while low-cloud cover decreased by up to 15.43%, increasing downward surface shortwave radiation by 64.09 W m^{−2}. Subsurface warming remained confined to the upper continental-shelf waters south of approximately 73° S, suggesting that local atmosphere-ice-radiation interactions contributed more strongly to the event than heat transport associated with ocean circulation. These results identify a distinctly polar pathway through which atmospheric forcing can generate compound thermal and biological extremes, highlighting the importance of local sea-ice coupling in shaping future Antarctic coastal productivity.

1. Introduction

Marine heatwaves (MHWs), defined as discrete periods of anomalously warm ocean temperatures exceeding a local climatological threshold (Hobday *et al* 2016), have increased in frequency, intensity, and duration globally under anthropogenic climate change (Oliver *et al* 2018, 2021), with projections indicating a 16–23-fold increase in MHW days at 1.5 °C–2.0 °C of warming (Frölicher and Laufkötter

2018, Laufkötter *et al* 2020). In the Southern Ocean, after decades of gradual expansion, Antarctic sea ice underwent a sudden and sustained decline beginning in 2016, with the winter maximum reaching a record low in September 2023 (Parkinson 2019, Hobbs *et al* 2024). This transition, driven by the confluence of subpolar surface warming, enhanced basal melting, and anomalous atmospheric circulation associated with a strengthened Amundsen Sea Low (Meehl *et al* 2019, Purich and Doddridge 2023, Wang *et al* 2023),

is increasingly recognized as a regime shift to a new low-extent state (Kusahara and Tatebe 2025). The resulting expansion of seasonally ice-free waters exposes vast areas of the Southern Ocean surface to enhanced atmospheric forcing, creating conditions conducive to prolonged MHW development in regions where such events were previously precluded by sea ice cover.

Satellite-era analyses indicate that Southern Ocean MHWs can enhance net primary production at high latitudes through increased water column stabilization and light availability (Montie *et al* 2020, Fernández-Barba *et al* 2024), with ecological consequences that differ markedly from the productivity suppression typically observed during lower-latitude MHWs (Smale *et al* 2019, Smith *et al* 2023). Despite these regional insights, tracing the precise sequence of events—from atmospheric triggers through ice-melt to the ultimate phytoplankton bloom—during a specific coastal extreme event remains challenging and largely unquantified. A key distinction of polar MHWs is that their drivers are mediated by coupled atmosphere-ice-ocean feedbacks rather than the oceanic advection and boundary current variability that dominate at lower latitudes (Holbrook *et al* 2019, 2020), yet this atmosphere-driven pathway has not been systematically documented for any individual Antarctic MHW event.

Resolving the event-scale atmosphere-ice-biology cascade requires a region where coastal polynyas, extreme thermal anomalies, and high biological sensitivity co-occur—conditions that converge in the Ross Sea. Despite occupying only 20% of Antarctica's ice-free continental shelf, the Ross Sea contributes approximately 28% of the Southern Ocean's total primary productivity (Arrigo *et al* 1998, 2008, Jones and Smith 2017), concentrated in coastal polynyas where early ice retreat triggers massive phytoplankton blooms sustained by iron supply from shelf sediments (Arrigo and Van Dijken 2003, Dinniman *et al* 2003, Arrigo *et al* 2015). Recent studies have advanced understanding of the climatological relationships governing this productivity: regime-based analyses have shown that the post-2016 sea ice decline shifted bloom phenology, with earlier ice melt advancing bloom timing though explaining only 10%–12.5% of chlorophyll-*a* (Chl-*a*) variability (Yang *et al* 2026), while spatiotemporal mapping of individual polynyas has revealed that moderate sea surface temperature (SST) and offshore wind conditions jointly regulate bloom intensity and spatial extent within the Ross Sea polynya system (Cha *et al* 2025). These statistical and correlative approaches have established the environmental covariates of Ross Sea productivity but have not addressed how an individual extreme event (e.g. MHW) propagates through the ice-ocean-biology system or identified the atmospheric mechanisms that initiate such events. The 2022–2023 austral summer, during which the eastern Ross Sea experienced

unprecedented thermal anomalies while the western sector retained persistent ice cover, presents a rare natural experiment to resolve these questions.

Here we combine satellite SST, ocean colour, sea ice, and atmospheric and oceanic reanalysis data to: (1) characterize the intensity, duration, and spatial extent of this extreme event; (2) quantify the temporal coupling among ice retreat, thermal forcing, and Chl-*a* response; and (3) identify the atmospheric circulation anomalies that initiated and sustained the compound MHW. The results trace, within a single basin and season, the full cascade from wind-driven polynya expansion through radiative amplification to a record phytoplankton bloom—a mechanistic sequence that we here confirm at the scale of an individual event, rather than infer from climatological composites.

2. Data and methods

2.1. Data

This study employs multiple reanalysis and satellite observational datasets to characterize the 2022–2023 Ross Sea MHW and associated ecological anomalies.

We use NOAA OISST v2.1 (0.25°, daily, 1981–present; Huang *et al* 2021) for MHW detection, ECMWF ORAS5 (0.25°, 75 vertical levels; Zuo *et al* 2019) for the upper-ocean (0–200 m) thermal structure averaged over 135° W–160° W, NOAA/NSIDC CDR v4 passive microwave sea ice concentration (SIC) (25 km, daily; Meier *et al* 2021) conservatively regridded to 0.25°, and ERA5 reanalysis (0.25°, hourly; Hersbach *et al* 2020) for atmospheric variables including winds, cloud cover, and radiation fluxes. Monthly-averaged ERA5 fields are used for analysis.

Chl-*a* concentration is obtained from the ESA Ocean Colour Climate Change Initiative (OC-CCI; Sathyendranath *et al* 2019), which merges multi-sensor observations (SeaWiFS, MODIS-Aqua, MERIS, VIIRS) from 1997 to the present at ~4 km resolution. Satellite observations in polar regions are frequently incomplete because of persistent cloud cover and sea-ice contamination; therefore, only valid cloud- and ice-free pixels were retained for analysis.

2.2. Methods

2.2.1. MHW detection

MHWs are identified following the standard framework of Hobday *et al* (2016), Hobday *et al* (2018). To reduce the influence of sea-ice contamination and ice-edge interpolation in OISST, we apply an additional sea-ice mask based on the NSIDC daily SIC dataset. Specifically, grid cells with SIC > 15% are excluded from the MHW analysis, and MHW detection is performed only for open-water grid cells with SIC ≤ 15%. For each valid grid point, we calculate

the 90th percentile SST threshold with a climatological baseline period of 1982–2011. The threshold is computed using an 11 day moving window (± 5 d around the target calendar day) to account for seasonal variability. An MHW event is defined when SST exceeds this threshold for five or more consecutive days. We quantify MHW cumulative intensity as the cumulative SST anomaly ($^{\circ}\text{C}\cdot\text{days}$) relative to the climatological mean across all MHW days within the 2022–2023 austral summer. MHW severity is classified into categories following Hobday *et al*: Category I (moderate, $1\text{--}2\times$ the climatological 90th percentile exceedance), Category II (strong, $2\text{--}3\times$), Category III (severe, $3\text{--}4\times$), and Category IV (extreme, $>4\times$).

2.2.2. Study area

Three key analysis regions are defined to characterize the contrasting responses across the Ross Sea: the eastern Ross Sea ($135^{\circ}\text{W}\text{--}160^{\circ}\text{W}$, $73^{\circ}\text{S}\text{--}76^{\circ}\text{S}$), the Ross Ice Shelf front ($170^{\circ}\text{E}\text{--}160^{\circ}\text{W}$, $77.2^{\circ}\text{S}\text{--}78.5^{\circ}\text{S}$), and Terra Nova Bay ($164^{\circ}\text{E}\text{--}168^{\circ}\text{E}$, $75^{\circ}\text{S}\text{--}77^{\circ}\text{S}$). The eastern Ross Sea box is selected because it corresponds to the main region where the 2022/23 MHW developed and where the strongest coupled SST, SIC, and Chl-a anomalies were observed. Terra Nova Bay is included as a contrasting western Ross Sea domain, because this region remained strongly influenced by persistent sea-ice cover during the event, therefore highlights the pronounced east–west heterogeneity in surface ocean conditions. The Ross Ice Shelf front is also considered because enhanced Chl-a concentrations were observed near the ice-shelf front; however, this region may be affected by ice-shelf meltwater and associated physical–biogeochemical processes that are not directly examined in this study.

2.2.3. Anomaly calculation and significance testing

The period 1982–2011 was selected as the climatological reference period for SST and atmospheric variables, and 1997–2016 for Chl-a; the latter reflects the onset of continuous multi-sensor ocean colour observations in late 1997, whereas physical variables permit standard 30 year climate normals. SST and Chl-a anomalies are computed as the difference between 2022–2023 observations and the respective climatological means. Statistical significance of anomalies is assessed using a two-standard-deviation criterion, with standard deviations calculated from the corresponding baseline periods. Trend significance is assessed using the non-parametric Mann–Kendall test, with trend magnitudes estimated using Sen's slope. Lag-1 autocorrelation is tested for each annual time series before trend analysis.

2.2.4. Shortwave cloud radiation effect (SWCRE)

The SWCRE quantifies the modulation of surface solar radiation by clouds and atmospheric processes,

defined as the difference between clear-sky and all-sky conditions:

$$\text{SWCRE} = \text{SSRD}_{\text{all_sky}} - \text{SSRD}_{\text{clear_sky}} \quad (1)$$

where $\text{SSRD}_{\text{all_sky}}$ and $\text{SSRD}_{\text{clear_sky}}$ are the downward surface shortwave radiation under all-sky and clear-sky conditions, respectively, from ERA5 reanalysis.

To quantitatively assess whether the enhanced shortwave radiation could account for the observed SST warming, we performed a simplified mixed-layer heat budget analysis. The monthly SST tendency was decomposed into the contributions from surface heat fluxes and a residual term:

$$\frac{\partial \text{SST}}{\partial t} = Q_{\text{SW}} + Q_{\text{LW}} + Q_{\text{LH}} + Q_{\text{SH}} + \text{Residual} \quad (2)$$

where Q_{SW} , Q_{LW} , Q_{LH} and Q_{SH} represent the mixed-layer temperature tendencies induced by shortwave radiation, longwave radiation, latent heat flux, and sensible heat flux, respectively. The residual term includes unresolved oceanic and sea-ice processes, as well as uncertainties in surface fluxes and mixed-layer depth.

The contribution of each surface heat-flux component was converted into a mixed-layer temperature tendency using:

$$Q_i = \frac{F_i}{\rho C_p h} \Delta t \quad (3)$$

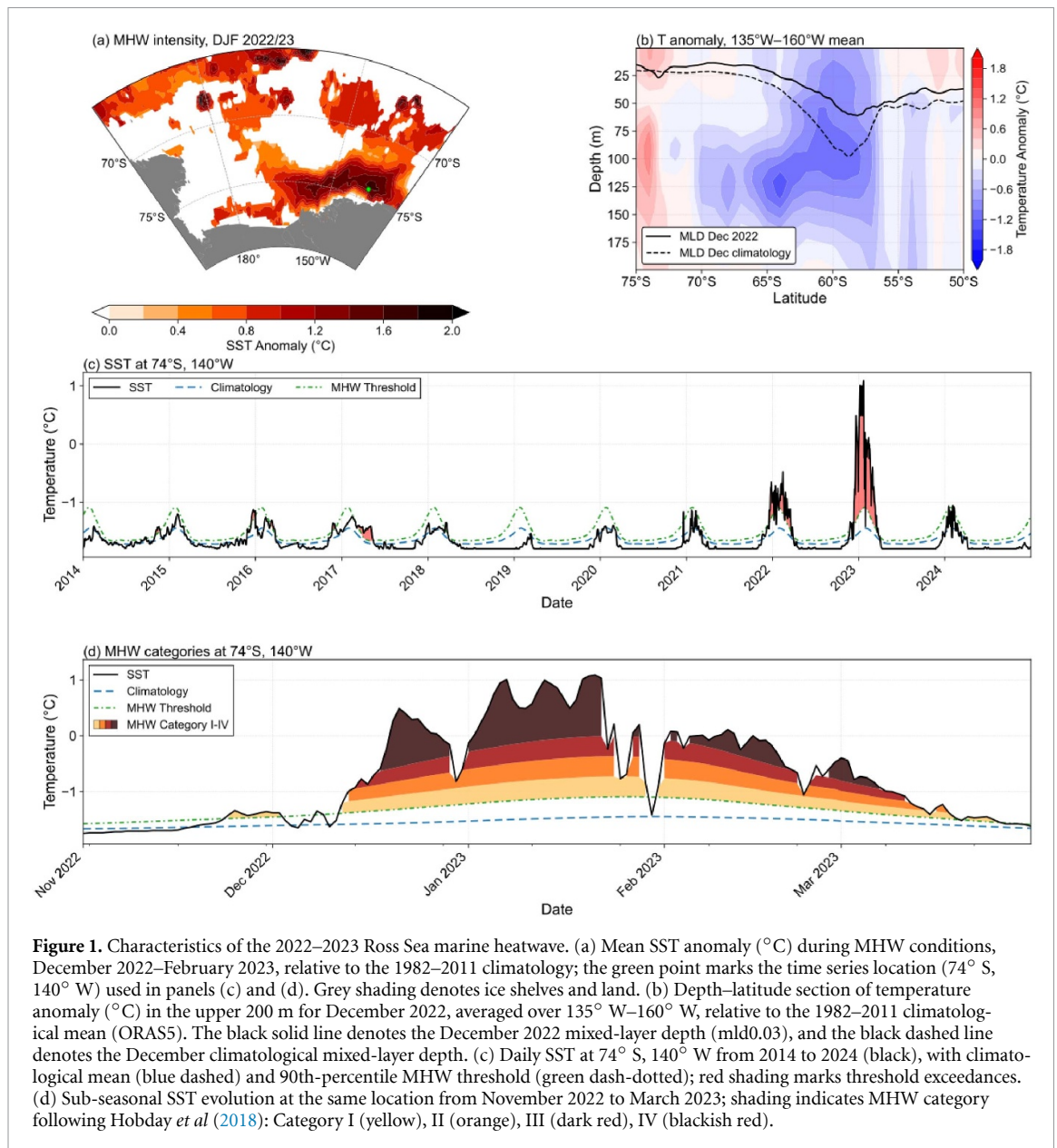
where F_i is the monthly mean surface heat flux, $\rho = 1025 \text{ kg m}^{-3}$, $C_p = 3990 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$, h is the monthly ORAS5 mixed-layer depth, and, Δt is the number of seconds in each month. ERA5 fluxes are converted to W m^{-2} and then to temperature contributions in $^{\circ}\text{C month}$, with positive values indicating mixed-layer warming.

The austral summer season is defined as December–February (DJF), spanning December 2022 to February 2023 for the 2022–2023 event.

3. Results

3.1. Spatiotemporal evolution of MHW associated Chl-a anomalies

The 2022–2023 austral summer MHW exhibits exceptional intensity and duration across the eastern Ross Sea (figure 1). The spatial distribution of MHW intensity (figure 1(a), calculated as mean SST anomaly during this summer) shows peak values exceeding $+2.0^{\circ}\text{C}$ concentrated near 74°S , 140°W (marked by green point), with the most intense warming confined to the eastern sector between 160°W and 135°W . This contrasts sharply with the western Ross Sea and the Terra Nova Bay region, where SST remains at or below climatological values and no MHW conditions occur, establishing a pronounced east–west thermal asymmetry across the Ross Sea.



Subsurface ocean temperature analysis is based on a longitude-averaged ORAS5 section over 135°W – 160°W for December 2022, together with the corresponding mld0.03 mixed-layer depth (figure 1(b)). December is selected because it exhibited the most rapid SST warming during the event. Between 73°S and 75°S , the upper 200 m shows widespread warming, with two pronounced warming cores located in the upper 10 m and at depths of 75–125 m, where temperature anomalies exceeded $+0.6^{\circ}\text{C}$. North of 73°S , temperature anomalies weaken rapidly and transition to near-zero or negative values throughout the 0–200 m depth range between 73°S and 55°S , indicating that the warming is regionally confined to the Ross Sea continental shelf (south of $\sim 73^{\circ}\text{S}$) rather than reflecting broader circumpolar ocean conditions.

Time series analysis at the representative location of maximum MHW intensity (74°S , 140°W)

characterizes the exceptional nature of the event in both historical and seasonal contexts (figures 1(c) and (d)). Over the decade 2014–2024 (figure 1(c)), daily SST at this location exhibits strong seasonal variability, remaining at the freezing point during autumn and winter due to sea ice presence, and rising above the freezing point during austral summer (December–February) but typically not exceeding -1°C . The 2022–2023 summer stands out dramatically: SST exceeds the 90th percentile MHW threshold (green dash-dotted line) for 105 cumulative days, the longest duration observed in the 10 year record. Peak SST reaches approximately 1.0°C in mid-January 2023, representing a $+2.0^{\circ}\text{C}$ anomaly relative to the climatological mean and the warmest summer conditions recorded at this location since 2000.

The sub-seasonal evolution during the 2022–2023 event (figure 1(d)) reveals a two-phase development. MHW conditions first appear briefly in

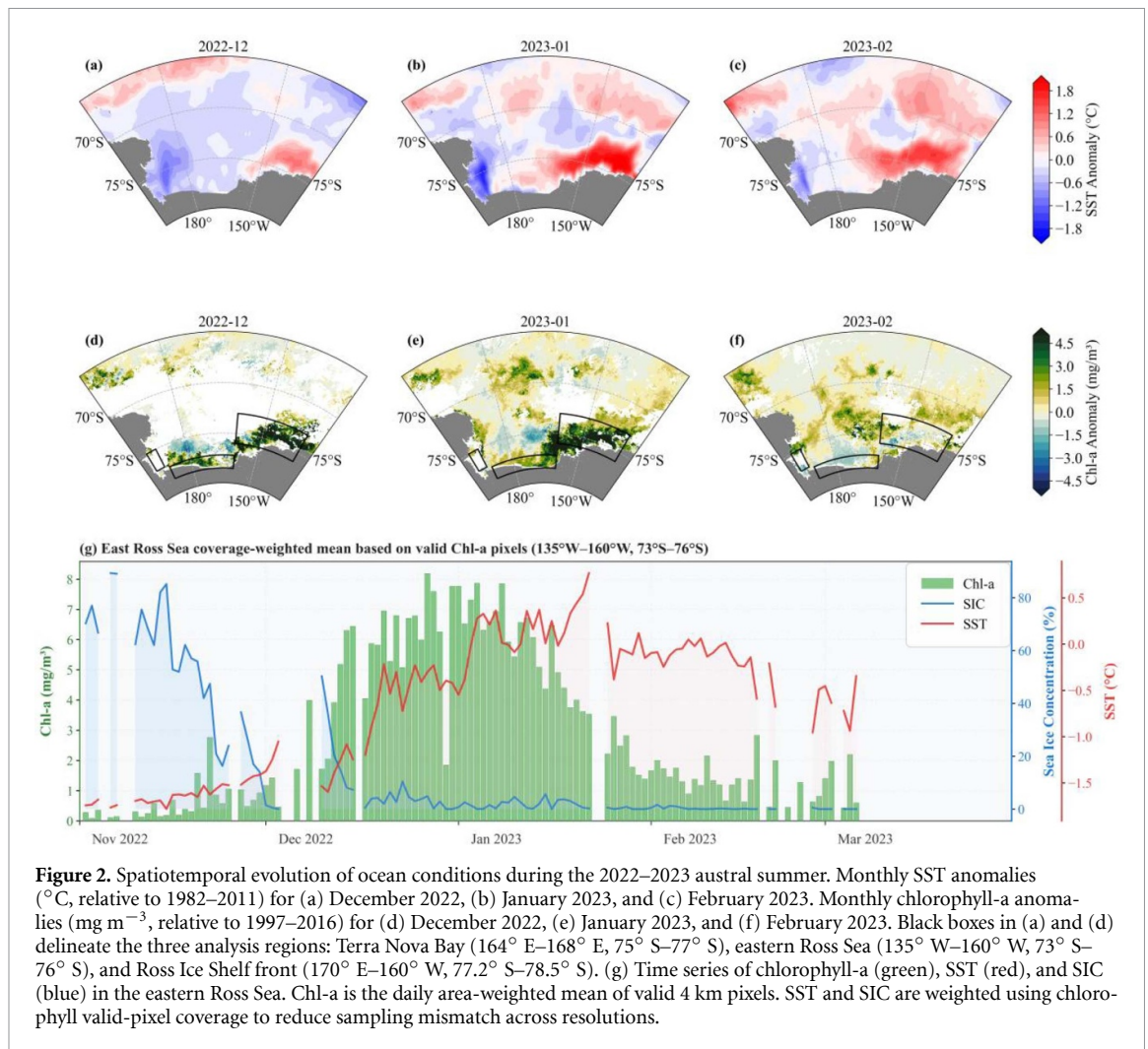


Figure 2. Spatiotemporal evolution of ocean conditions during the 2022–2023 austral summer. Monthly SST anomalies (°C, relative to 1982–2011) for (a) December 2022, (b) January 2023, and (c) February 2023. Monthly chlorophyll-a anomalies (mg m⁻³, relative to 1997–2016) for (d) December 2022, (e) January 2023, and (f) February 2023. Black boxes in (a) and (d) delineate the three analysis regions: Terra Nova Bay (164° E–168° E, 75° S–77° S), eastern Ross Sea (135° W–160° W, 73° S–76° S), and Ross Ice Shelf front (170° E–160° W, 77.2° S–78.5° S). (g) Time series of chlorophyll-a (green), SST (red), and SIC (blue) in the eastern Ross Sea. Chl-a is the daily area-weighted mean of valid 4 km pixels. SST and SIC are weighted using chlorophyll valid-pixel coverage to reduce sampling mismatch across resolutions.

late November 2022 at weak intensity but quickly subside. The event re-emerges in mid-December 2022 and undergoes explosive intensification, escalating from moderate to extreme within approximately two weeks by late December. The event maintains Category IV intensity throughout mid-January 2023, when SST anomalies exceed four times the baseline 90th percentile threshold, before gradually weakening through February and returning to near-normal conditions by late March 2023. This progression demonstrates sustained atmospheric and oceanic forcing throughout the austral summer, rather than a brief episodic warming event. The cumulative MHW intensity reaches 127.22 °C days in January 2023, more than double any previous event recorded at this location since 2000 (see section 3.2 for historical context).

SIC averaged over the eastern Ross Sea (figure 2(g)) decreased rapidly from approximately 80% to below 15% in late November 2022 as the polynya expanded, producing widespread open-water conditions earlier than the climatological seasonal cycle. Chl-a subsequently increased during December and peaked in late December–early January, while SST increased and remained

anomalously high. During January–February, Chl-a gradually declined as the bloom weakened, whereas SST remained elevated. This decoupling indicates that the persistent SST anomaly was not closely tied to contemporaneous changes in phytoplankton biomass. We therefore interpret the warm SST primarily as part of a favourable physical background associated with reduced sea ice, expanded open water, and enhanced surface heating, rather than as evidence that warming directly drove the bloom or that biological production regulated the upper-ocean thermal state. This temporal sequence—ice retreat in late November, followed by SST warming in early-to-mid December, and culminating in the Chl-a peak in late December and early January—indicates the sequential removal of limiting factors: first, light availability through ice loss, then temperature-enhanced growth rates through MHW development.

3.2. Historical context and interannual variability

To assess whether the 2022–2023 conditions were unprecedented, we examine the interannual variability and long-term trends of key ocean-ice-biology variables in the eastern Ross Sea over the past two decades (figures 3(a)–(c)).

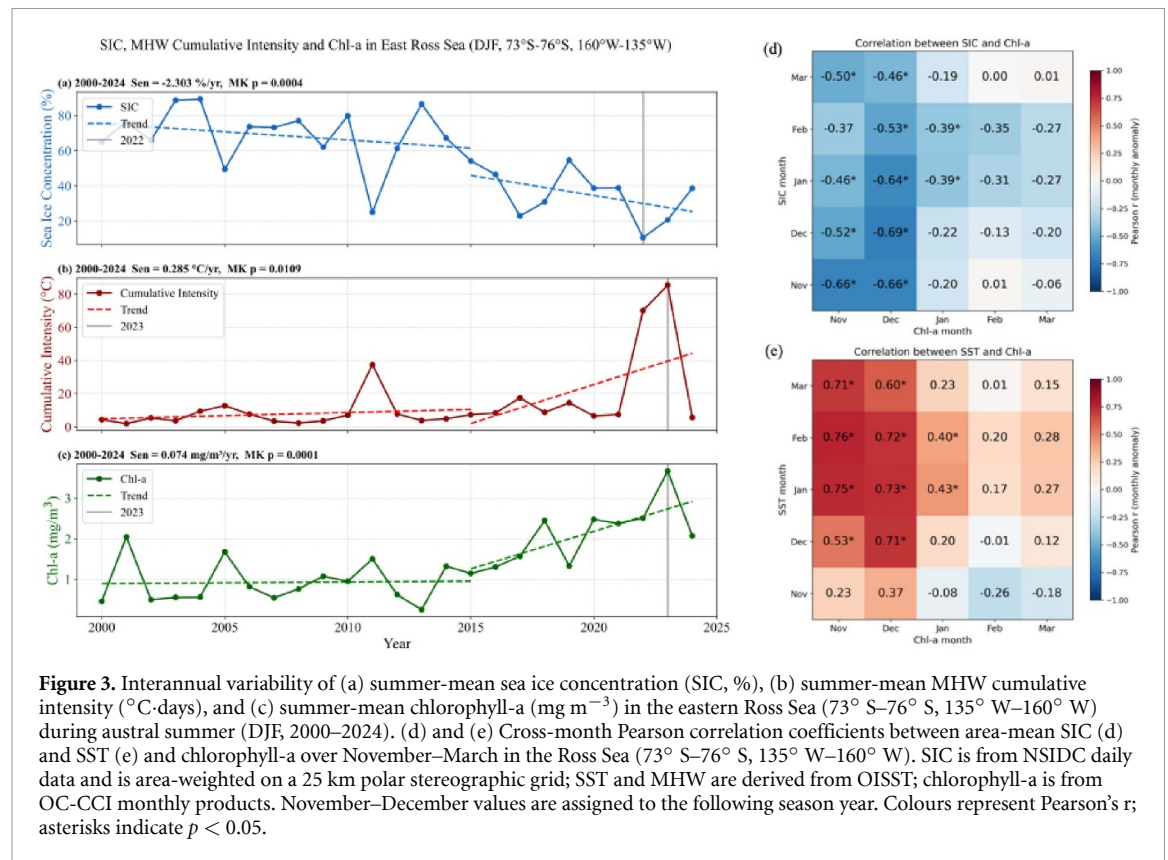


Figure 3. Interannual variability of (a) summer-mean sea ice concentration (SIC, %), (b) summer-mean MHW cumulative intensity (°C-days), and (c) summer-mean chlorophyll-a (mg m^{-3}) in the eastern Ross Sea (73°S – 76°S , 135°W – 160°W) during austral summer (DJF, 2000–2024). (d) and (e) Cross-month Pearson correlation coefficients between area-mean SIC (d) and SST (e) and chlorophyll-a over November–March in the Ross Sea (73°S – 76°S , 135°W – 160°W). SIC is from NSIDC daily data and is area-weighted on a 25 km polar stereographic grid; SST and MHW are derived from OISST; chlorophyll-a is from OC-CCI monthly products. November–December values are assigned to the following season year. Colours represent Pearson's r ; asterisks indicate $p < 0.05$.

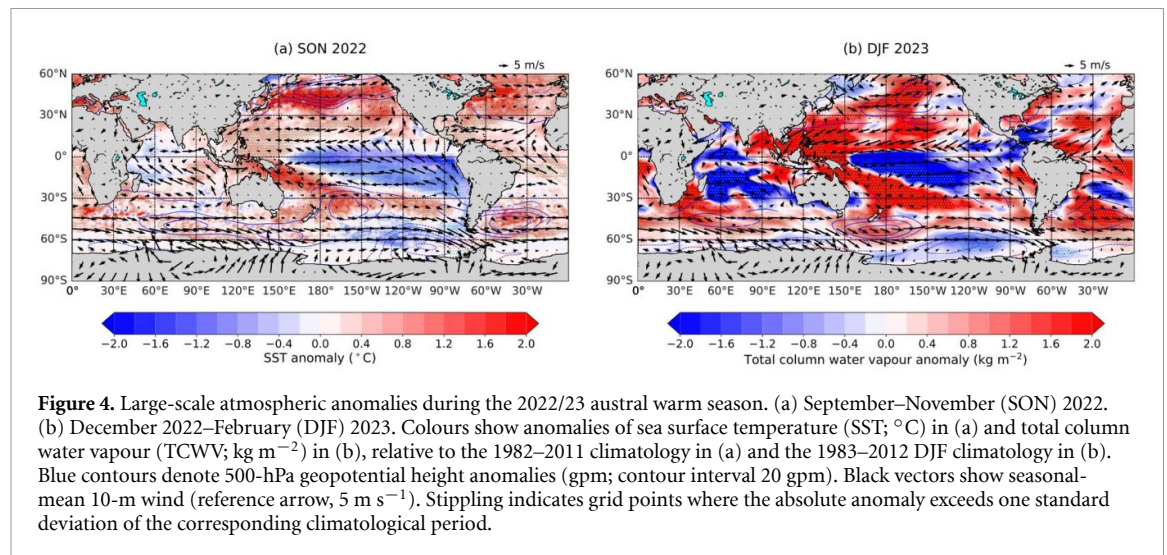
The observed decline in sea ice in the eastern Ross Sea must be considered within the broader context of Antarctic sea ice regime change. Antarctic sea ice extent showed a gradual increase from the late 1970s to 2015, followed by an abrupt and sustained decline beginning in late 2016. This transition has been widely interpreted as a potential regime shift to a new low-extent state, driven by the combined effects of Southern Ocean subsurface warming and atmospheric circulation anomalies (Purich and Doddridge 2023, Kusahara and Tatebe 2025).

Daily SST evolution at 74°S , 140°W for 2014–2023 (figure 1(c)) reveals that the 2022–2023 summer experienced sustained and intense warming far exceeding previous events. While earlier MHWs occurred (e.g. 2021–2022: 65 d duration, 74°C days cumulative intensity), the 2022–2023 event was more than twice as long and intense, making it the most extreme thermal anomaly in the regional satellite record. Using ERA5-derived surface shortwave radiation converted to photosynthetically active radiation (PAR; assuming $\text{PAR} \approx 45\%$ of shortwave radiation; Frouin and Pinker 1995), we estimate that photosynthetically available radiation increased by 30% relative to the climatological mean, substantially enhancing light availability for phytoplankton growth.

Notably, Chl-a variability in the eastern Ross Sea appears to share a similar temporal inflection. The 2022–2023 season exhibits the highest summer-mean Chl-a concentration (3.67 mg m^{-3}) since satellite records began in 1997 (figure 3(c)), exceeding the

long-term mean (1.29 mg m^{-3}) by more than 2 standard deviations. While a modest positive trend is evident over the full record, this trend is largely concentrated in the post-2016 period, coinciding with the accelerated sea ice decline. The substantial interannual variability (coefficient of variation = 55.09%) indicates that factors beyond the linear ice trend, including MHW occurrence, cloud cover anomalies, and wind-driven nutrient supply—modulate productivity on year-to-year timescales. Nevertheless, the co-occurrence of intensified MHW activity, reduced sea ice, and elevated Chl-a since 2016 suggests that the ongoing sea ice regime shift may be creating conditions increasingly favourable for extreme biological productivity events in the eastern Ross Sea.

Regional lag-correlation analysis of area-averaged monthly anomalies in the eastern Ross Sea further shows a robust negative association between SIC and Chl-a during November–March (figures 3(d) and (e)). Eleven of the 25 month-pair correlations are significant ($p < 0.05$), with the strongest correlation between December SIC and December Chl-a ($r = -0.69$). The simultaneous and lagged correlations suggest that reduced SIC likely preconditioned phytoplankton growth, with an apparent response time of approximately 0–1 months. This timescale is consistent with Antarctic phytoplankton doubling times of several days to approximately one week (Eppeley 1972, Smith and Gordon 1997), together with the time required for ice clearance, upper-ocean restratification, and initial bloom establishment



(Li *et al* 2016, Xu *et al* 2024). Although SST was generally positively correlated with Chl-a, several strong correlations involved SST anomalies occurring after the corresponding Chl-a anomalies. Thus, low SIC and warm surface conditions likely favoured the bloom, but their effects cannot be isolated from unobserved controls such as nutrients, iron supply, mixed-layer depth, and grazing.

3.3. Physical drivers of the 2022–2023 MHW

3.3.1. Coastal polynya expansion and large-scale climate variability

The anomalous ice loss in the eastern Ross Sea is initiated by enhanced offshore (southerly) wind anomalies that appear as early as October–November 2022, before the austral summer peak (supplementary figure 1; figure 4). In November, meridional wind anomalies show a pronounced east-west dipole across the Ross Sea: enhanced southerly flow over the eastern sector drives sea ice northward away from the coast, while anomalous northerly flow over the western sector (including Terra Nova Bay) transports ice toward the coast and maintains ice coverage (supplementary figure 2(c)). Although the strongest meridional wind anomaly (+2.13 m s⁻¹) occurred in October 2022, before the polynya formed (supplementary figure 2(b)), sustained southerly anomalies throughout the summer prevented sea-ice re-advance and maintained the ice-free conditions necessary for the prolonged MHW and associated biological response.

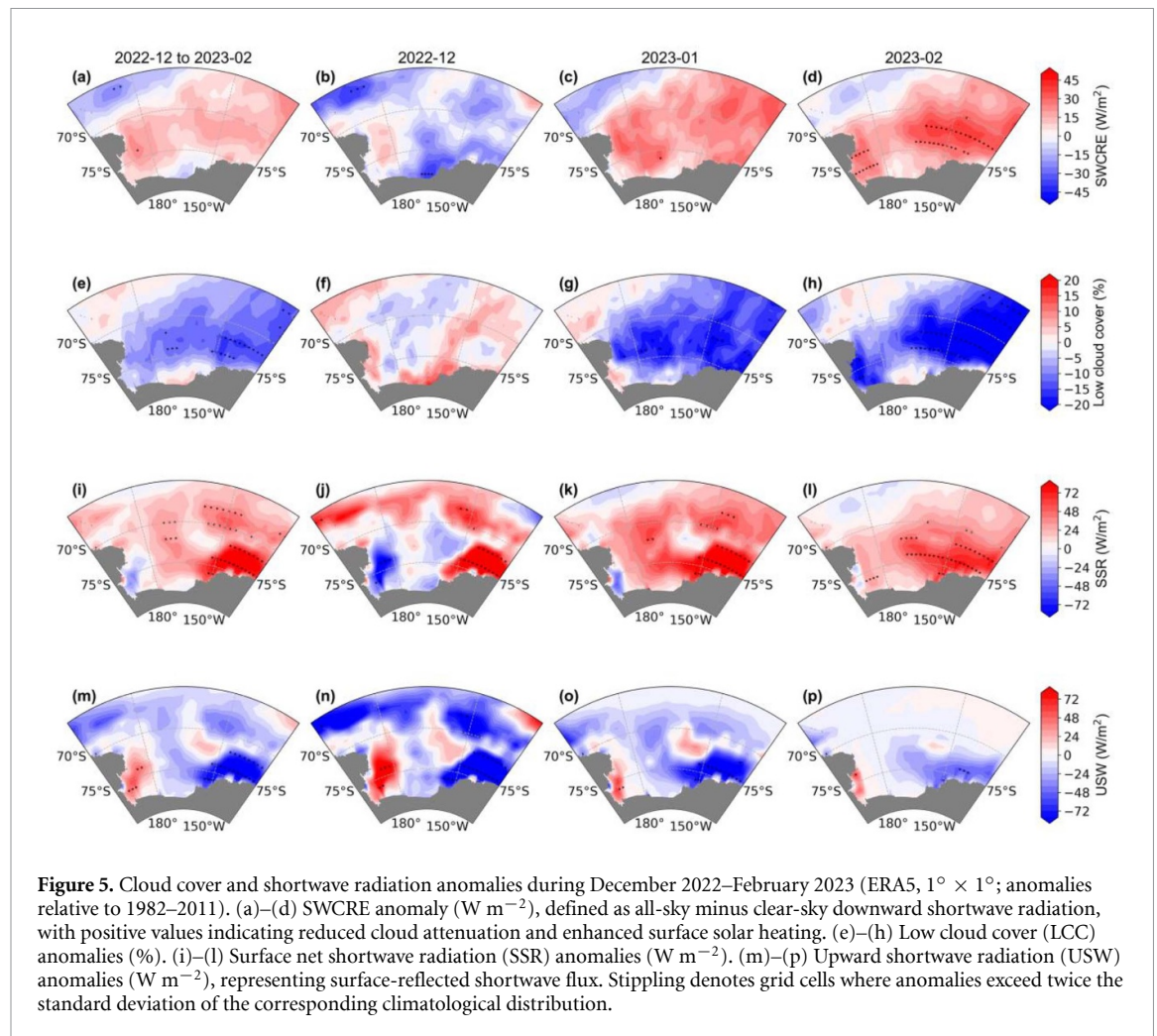
During SON 2022, the tropical Pacific experienced pronounced cooling associated with La Niña conditions, while the southern annular mode remained predominantly in its positive phase (figure 4(a)). During DJF 2023, total column water vapour over the Ross Sea was markedly reduced (figure 4(b)). The reduced atmospheric moisture content was consistent with the observed decrease in low cloud cover (figure 5(e)), favouring enhanced

surface net shortwave radiation (SSR), accelerated sea-ice retreat, and subsequent surface warming. These results suggest that large-scale climate variability provided a favourable background for the regional atmosphere–sea–ice–ocean processes responsible for the 2022–2023 MHW and phytoplankton bloom, rather than directly driving the event.

3.3.2. Local atmospheric forcing: clouds and radiation

The pronounced warming in the eastern Ross Sea is driven in part by anomalous atmospheric conditions that enhance surface solar radiation (figure 5). Low cloud cover anomalies during the DJF period (figure 5(e)) show reduced cloudiness over the eastern Ross Sea, with the monthly evolution indicating the strongest negative anomalies (−15.43%) occurring in January 2023 (figure 5(g)) coinciding with the peak MHW intensity. SSR anomalies (figures 5(i)–(l)) confirm enhanced solar heating, with positive anomalies of +64.09 W m⁻² over the eastern polynya region during January 2023.

The Shortwave Cloud Radiation Effect (SWCRE; figures 5(a)–(d)) quantifies the cloud contribution: positive SWCRE anomalies of +22.90 W m⁻² indicate a weakening of the climatological cloud shading effect, allowing substantially more shortwave radiation to reach the ocean surface, directly contributing to SST warming. The strong spatial correspondence between low-cloud-cover and SWCRE anomalies (supplementary figure 3) suggests that reduced low-cloud cover was a major contributor to the weakened cloud-shading effect. Meanwhile, the approximate mixed-layer heat-budget analysis (supplementary figure 4) shows that shortwave radiation provided the largest positive contribution to the mixed-layer warming, substantially exceeding the contributions from the other surface heat-flux components. Together, these results indicate that enhanced shortwave absorption, arising from both reduced cloud



cover and reduced sea-ice albedo, played the leading role in the observed warming.

4. Discussion

4.1. Atmosphere-ice-radiation feedback as the driver

The 2022–2023 eastern Ross Sea MHW developed after a sequence of atmospheric, sea-ice, and radiative anomalies that together maintained elevated SST over the continental shelf for more than three months. A meridional wind dipole drove divergent ice transport and the early expansion of the eastern Ross Sea polynya, while low-cloud cover decreased by up to 15.43%, increasing downward surface shortwave radiation by 64.09 W m^{-2} . The synergistic effect of expanded ice-free area and intensified solar heating sustained elevated SST for over three months. Subsurface temperature analysis indicates that the warming anomaly was primarily confined to the upper ocean south of 73° S , with the strongest signal occurring within the upper 10 m and 75–125 m. In contrast to MHWs in western boundary current systems such as the Kuroshio and the California Current, where subsurface warming

associated with warm-water advection can often extend through much of the upper 300 m (Holbrook *et al* 2019, 2020), no pronounced subsurface warming was evident below 150 m in the 73° S – 55° S depth section (figure 1(b)). Together with the mixed-layer heat-budget analysis, these results suggest that enhanced shortwave radiation was the primary driver of the 2022–2023 MHW, although contributions from ocean circulation cannot be excluded. In the Ross Sea, sea ice introduces a feedback absent at lower latitudes: ice removal simultaneously enhances shortwave absorption and eliminates the insulating barrier between ocean and atmosphere, compounding the warming beyond what cloud–radiation interactions alone can produce (Amaya *et al* 2020).

4.2. Threshold-dependent ecological response and spatial heterogeneity

Unlike lower-latitude MHWs, which generally suppress primary productivity by enhancing stratification in nutrient-depleted waters (Smale *et al* 2019, Smith *et al* 2023), the 2022–2023 Ross Sea event drove record phytoplankton productivity, consistent with recent findings of enhanced production during high-latitude MHWs (Montie *et al* 2020, Fernández-Barba *et al* 2024). Critically, the ecological response

was threshold-dependent rather than linearly proportional to warming. Ice retreat in late November 2022 first eliminated light limitation, but substantial Chl-*a* increases occurred only after an approximately 2–3 week lag required for restratification and bloom establishment. Once both light and thermal thresholds were exceeded, Chl-*a* tracked SST fluctuations with near-synchronous co-variation. The contrast with Terra Nova Bay, where persistent ice cover meant neither threshold was met and no comparable bloom developed, supports that the extreme bloom required the synergistic co-occurrence of early ice retreat and sustained MHW intensity. This east–west asymmetry within a single basin demonstrates that spatial heterogeneity in ice–atmosphere coupling can generate sharply contrasting ecological outcomes over short distances—a complexity that reflects atmosphere–ice–biology coupling operating on event timescales, complementing the climatological ice–productivity relationships established by prior studies.

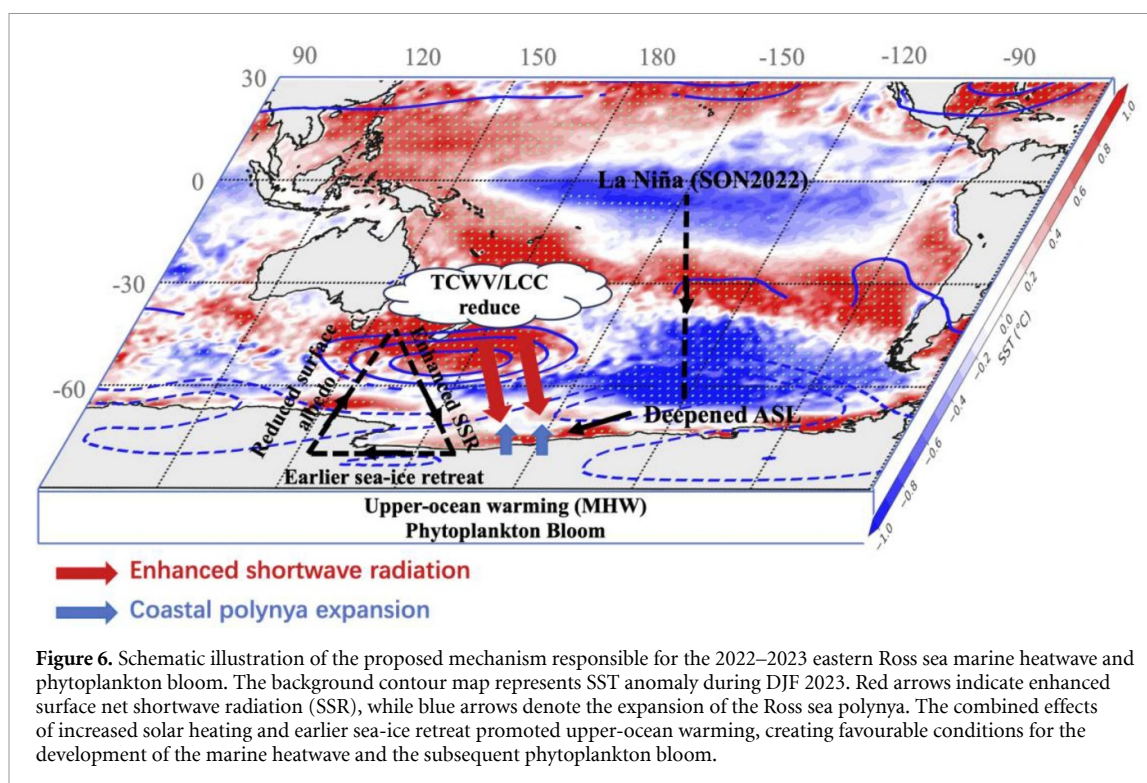
4.3. Limitations

Our analysis relies on ERA5 reanalysis for atmospheric variables in a data-sparse region. ERA5 overestimates downwelling shortwave radiation over the Southern Ocean, with mean biases of $\sim 54 \text{ W m}^{-2}$ during austral summer south of the Antarctic Polar Front (Mallet *et al* 2023), linked to underrepresentation of multi-layer cloud structures (Bodas-Salcedo *et al* 2014, Truong *et al* 2022). Satellite Chl-*a* retrievals suffer from extensive data gaps in polar regions due to cloud obstruction, sea ice, and low solar elevation (Park *et al* 2020, Hong *et al* 2023), with valid observation ratios often below 30% at high southern latitudes. In this study, regional Chl-*a* was calculated as an area-weighted mean using all valid pixels within each daily period, which reduces the influence of uneven spatial sampling. Nevertheless, variations in valid-pixel coverage may still introduce uncertainty into the estimated SST–Chl-*a* relationship. The lack of concurrent nutrient and iron observations means we cannot rule out a contribution from enhanced

iron supply via intensified ice melt or wind-driven upwelling. Therefore, our study focused on the eastern Ross Sea polynya rather than the Ross Ice Shelf front, where meltwater-derived iron supply may exert additional control on productivity. Finally, this study documents a single event; whether the atmosphere-driven pathway identified here generalizes to other polar MHWs or is specific to the 2022–2023 atmospheric configuration requires investigation across additional events and through targeted modelling.

5. Conclusions

We document a Category IV MHW in the eastern Ross Sea during the 2022–2023 austral summer, the most intense thermal event in the regional satellite record, and trace within a single basin and season the full cascade from atmospheric forcing to a record phytoplankton bloom. A meridional wind dipole expanded the eastern Ross Sea polynya from late November 2022, while anomalously low cloud cover enhanced surface shortwave radiation. The resulting ice albedo and cloud radiation forcing, rather than oceanic heat advection, sustained SST anomalies of up to approximately $+2.5 \text{ }^{\circ}\text{C}$ for more than three months, with warming confined to shelf waters south of $\sim 73^{\circ} \text{ S}$. Summer-mean Chl-*a* reached 3.67 mg m^{-3} , the highest since ocean-colour records began in 1997. The biological response was threshold-dependent rather than proportional to warming, requiring the co-occurrence of early ice retreat and sustained thermal anomalies. No comparable bloom developed in the ice-covered western sector, where that combination was absent. This atmosphere-driven pathway is mechanistically distinct from the MHWs of lower latitudes (figure 6). Priority observations to further constrain these processes include biogeochemical Argo profiling to resolve nutrient dynamics during MHW events and sustained radiation monitoring over the Ross Sea to validate reanalysis-derived forcing.



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Data availability statement

All the data analysed here are openly available. The satellite-retrieved sea ice concentration is obtained from the National Snow and Ice Data Center (NSIDC; <https://nsidc.org/>). Both ERA5 and ORAS5 are retrieved from the Copernicus Climate Data Store (<https://cds.climate.copernicus.eu/>). The daily Optimum Interpolation Sea Surface Temperature (OISST v2.1) is obtained from NOAA (www.ncei.noaa.gov/products/optimum-interpolation-sst). The ocean colour chlorophyll-a data are obtained from the ESA Ocean Colour Climate Change Initiative (OC-CCI; www.oceancolour.org/).

Supplementary data 1 available at: <https://doi.org/10.1088/1748-9326/ae9780/data1>.

Conflict of interest

We declare no competing interests.

Ethics statement

This study uses only publicly available data. No human or animal experiments were involved, so ethical approval was not required.

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